ESERCIZI GEOMETRIA delle AREE

Ripasso

$$\{S\} = \begin{Bmatrix} S_y \\ S_x \end{Bmatrix} = \begin{Bmatrix} \int_A x\,dA \\ \int_A y\,dA \end{Bmatrix}$$

Se l'ORIGINE è G (BARICENTRO) $\Rightarrow S_x, S_y = 0$ $\Rightarrow$ $x_G = \frac{S_y}{A}$, $y_G = \frac{S_x}{A}$

$$[I] = \begin{bmatrix} I_y & I_{xy} \\ I_{xy} & I_x \end{bmatrix} = \begin{bmatrix} \int x^2 dA & \int xy\,dA \\ \int xy\,dA & \int y^2 dA \end{bmatrix}$$

Se sistema di riferimento PRINCIPALE $\Leftrightarrow I_{xy} = 0$

$\rightarrow$ gli assi sono ASSI PRINCIPALI D'INERZIA

Se un sistema di riferimento è BARICENTRICO e PRINCIPALE

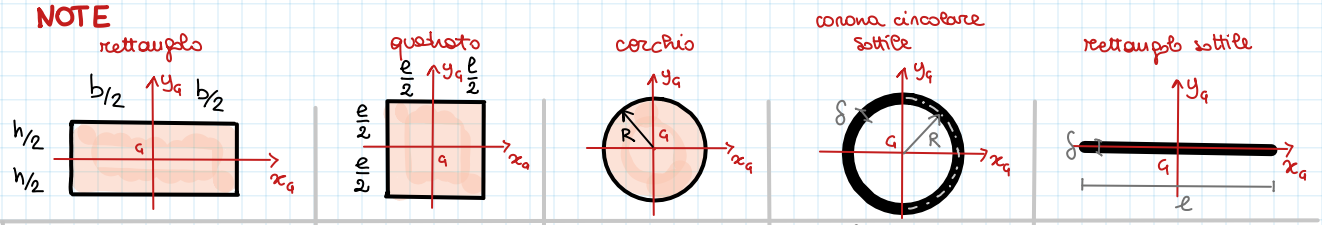
$\Downarrow$

CENTRALE

$S_x, S_y = 0$

$I_{xy} = 0$

SEZIONI NOTE

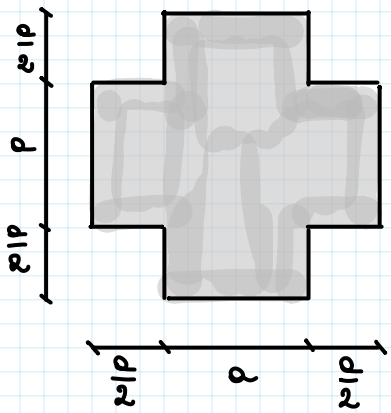


	rettangolo	quadrato	cerchio	corona circolare sottile	rettangolo sottile
				$\delta \ll R$	$\delta \ll e$
A	bh	e^2	πR^2	$\delta R 2\pi$	$e\delta$
S_x	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
S_y	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
I_x	$bh^3/12$	$e^4/12$	$\frac{\pi R^4}{4}$	$\delta R^3 \pi$	$\delta^3 e/12 \sim \emptyset$
I_y	$hb^3/12$	$e^4/12$	$\frac{\pi R^4}{4}$	$\delta R^3 \pi$	$e^3 \delta/12$
I_{xy}	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

sez. giroscopiche ($I_x = I_y$)

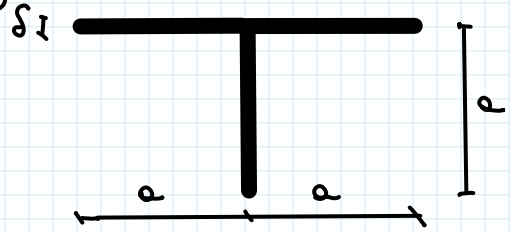
ESERCIZI - Calcolare i momenti d'inerzia assiali nel sistema di riferimento centrale

①



$a = 10$ mm

②

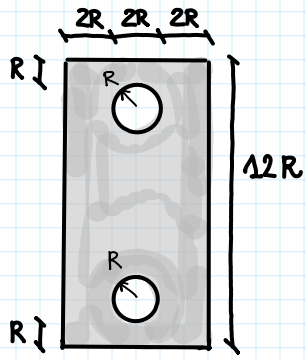


δ spessore costante

$\delta = 2$ mm

$a = 25$ mm

③



$R = 0.25$ mm